Manufacturing & Video-Analytics Intelligent Report

“I want the data from the mes application for machine utilization and from video analytics I want camera configuration information”

Generated: 2026-10-19 17:49:40
Time filter applied

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Executive Summary

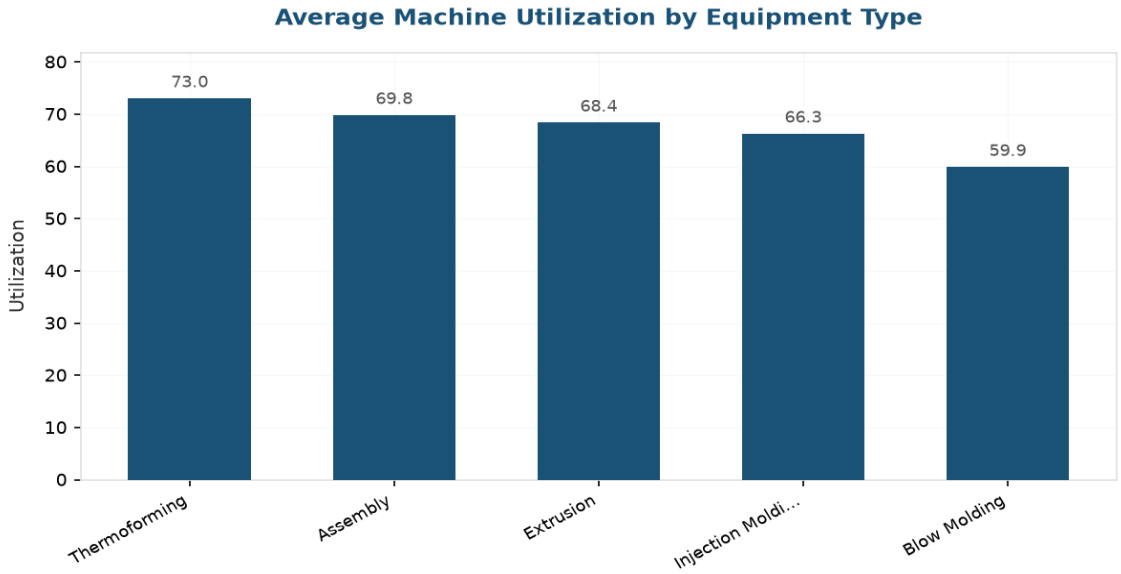
2	12	333	Applied	0
DATABASES QUERIED	TABLES MATCHED	TOTAL RECORDS	TIME FILTER	FAILED QUERIES

Matched keywords: camera, configuration, machine, utilization

Telemetry window: 2026-10-18T12:19:40Z → 2026-10-19T12:19:40Z (bucketed every 5m)

MES Machine Utilization and Capacity Analysis

Analysis of the MES MachineMaster database reveals 40 registered machines across 5 primary equipment categories including Injection Molding, Thermoforming, Assembly, Extrusion, and Blow Molding. Machine utilization ranges from a low of 43.40% to a maximum of 92.70%, with an overall plant average utilization rate of 68.61%. Production capacity per hour averages 216.60 units (ranging from 52.82 to 396.75 units/hr) alongside an average overall capacity of 541.73 units per machine. Shift-level granular monitoring from the CapacityAnalysis dataset across Machine IDs 1 through 4 shows an average utilization of 72.92%, with individual shifts hitting up to 100.00% capacity while others drop as low as 37.50%. These operational figures reflect moderate overall productivity across the shop floor but highlight scheduling imbalances and underutilized equipment running well below potential.



Data preview — MachineMaster (showing 10 of 40 points/rows)

Utilization	MachineCode	MachineId	MachineName	MachineType	Capacity	CapacityPerHour	CustomerId
81.90	M001	1	Injection Molding Line 1	Injection Molding	772.96	178.64	1
45.00	M002	2	Thermoforming Line 2	Thermoforming	554.94	77.34	1
58.90	M003	3	Thermoforming Line 3	Thermoforming	826.22	229.64	1
75.90	M004	4	Thermoforming Line 4	Thermoforming	681.29	150.48	1
59.00	M005	5	Assembly Line 5	Assembly	120.46	235.09	1
69.70	M006	6	Injection Molding Line 6	Injection Molding	190.81	333.32	1
43.40	M007	7	Extrusion Line 7	Extrusion	705.28	335.53	1
56.50	M008	8	Thermoforming Line 8	Thermoforming	572.02	66.41	1

Utilization	MachineCode	MachineId	MachineName	MachineType	Capacity	CapacityPerHour	CustomerId
77.10	M009	9	Assembly Line 9	Assembly	283.00	388.73	1
70.40	M010	10	Blow Molding Line 10	Blow Molding	943.44	217.79	1

Video Analytics Camera Configuration and Status Audit

The Video Analytics master table defines 3 configured camera devices: Flarehub (Camera 1), Luxsphere (Camera 2), and Entry Gate (Camera 6), all managed under the admin user account. The Entry Gate camera is assigned to Plant ID 1, Department ID 1, and Location ID 1, whereas Flarehub and Luxsphere lack specific facility mapping. Although all three cameras are listed as online in the primary cameras table, the camera_status_logs table records 50 consecutive offline status entries for camera IDs 1 through 3. This clear contradiction between configuration records and status logs indicates a communication timeout or data pipeline integration gap that requires technical audit.

Data preview — cameras (showing 3 of 3 points/rows)

camera_number	department_id	location_id	name	plant_id	status	user_id
1	None	None	Flarehub	None	online	admin
2	None	None	Luxsphere	None	online	admin
6	1	1	Entry Gate	1	online	admin

Appendix — Data Quality

Table	Records	Completeness	Outliers	Score
cameras	3	71.4%	0	Needs review
camera_status_logs	50	100.0%	0	Good
MachineMaster	40	100.0%	0	Good
CapacityAnalysis	50	100.0%	0	Good
Machine	40	100.0%	0	Good
MachineFGMapping	50	100.0%	0	Good
MachineMaterialMapping	50	100.0%	0	Good
WeighingMachine	50	100.0%	0	Good

LLM Usage & Cost

Model	Input tokens	Output tokens	Total tokens	Estimated cost
gemini/gemini-3.5-flash	5,372	2,598	7,970	$0.0314